Assignmet

Pivot Table & Charts

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Assignment Name:Pivot Table & Charts

Drive link:NA

Github link:NA

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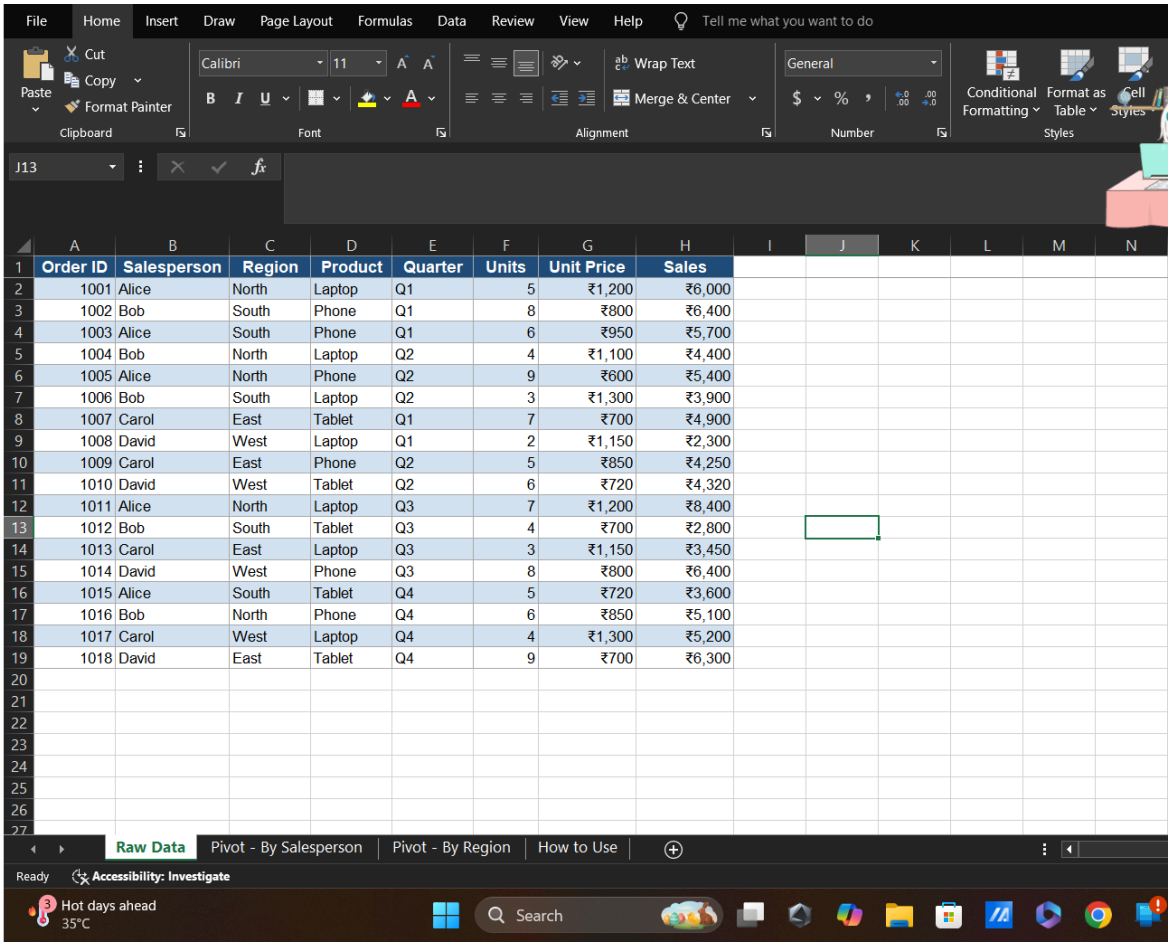
1. **What is the purpose of a PivotTable in Excel?**

Ans-

To use a Pivot Table in Excel, select your data range, click Insert > PivotTable, and drag fields into rows, columns, or values to summarize data instantly without formulas. It transforms large datasets into compact, interactive reports to show totals, averages, or counts by category.

Example:



Order ID	Salesperson	Region	Product	Quarter	Units	Unit Price	Sales
1001	Alice	North	Laptop	Q1	5	₹1,200	₹6,000
1002	Bob	South	Phone	Q1	8	₹800	₹6,400
1003	Alice	South	Phone	Q1	6	₹950	₹5,700
1004	Bob	North	Laptop	Q2	4	₹1,100	₹4,400
1005	Alice	North	Phone	Q2	9	₹600	₹5,400
1006	Bob	South	Laptop	Q2	3	₹1,300	₹3,900
1007	Carol	East	Tablet	Q1	7	₹700	₹4,900
1008	David	West	Laptop	Q1	2	₹1,150	₹2,300
1009	Carol	East	Phone	Q2	5	₹850	₹4,250
1010	David	West	Tablet	Q2	6	₹720	₹4,320
1011	Alice	North	Laptop	Q3	7	₹1,200	₹8,400
1012	Bob	South	Tablet	Q3	4	₹700	₹2,800
1013	Carol	East	Laptop	Q3	3	₹1,150	₹3,450
1014	David	West	Phone	Q3	8	₹800	₹6,400
1015	Alice	South	Tablet	Q4	5	₹720	₹3,600
1016	Bob	North	Phone	Q4	6	₹850	₹5,100
1017	Carol	West	Laptop	Q4	4	₹1,300	₹5,200
1018	David	East	Tablet	Q4	9	₹700	₹6,300

Core Purpose

It helps you **analyze large datasets quickly** by reorganizing ("pivoting") rows and columns to reveal patterns, comparisons, and insights.

Key Use Cases

- **Sales reports** — totals by region, rep, or product
- **Financial summaries** — monthly/quarterly expense breakdowns
- **HR analytics** — headcount by department or role
- **Inventory tracking** — stock levels by category
- **Survey analysis** — response counts by question/group

How to Create One (Quick Steps)

1. Click anywhere in your data
2. Go to **Insert** → **PivotTable**
3. Choose where to place it → click **OK**
4. Drag fields into **Rows**, **Columns**, and **Values**

2.Create a PivotTable to show Total Sales by Region.

Ans-

To create a PivotTable showing Total Sales by Region, select your data range, click Insert > PivotTable, then drag "Region" to the Rows area and "Sales" to the Values area. This instantly summarizes total sales per region, which can be further formatted or sorted by right-clicking the values

Detailed Steps to Create the PivotTable

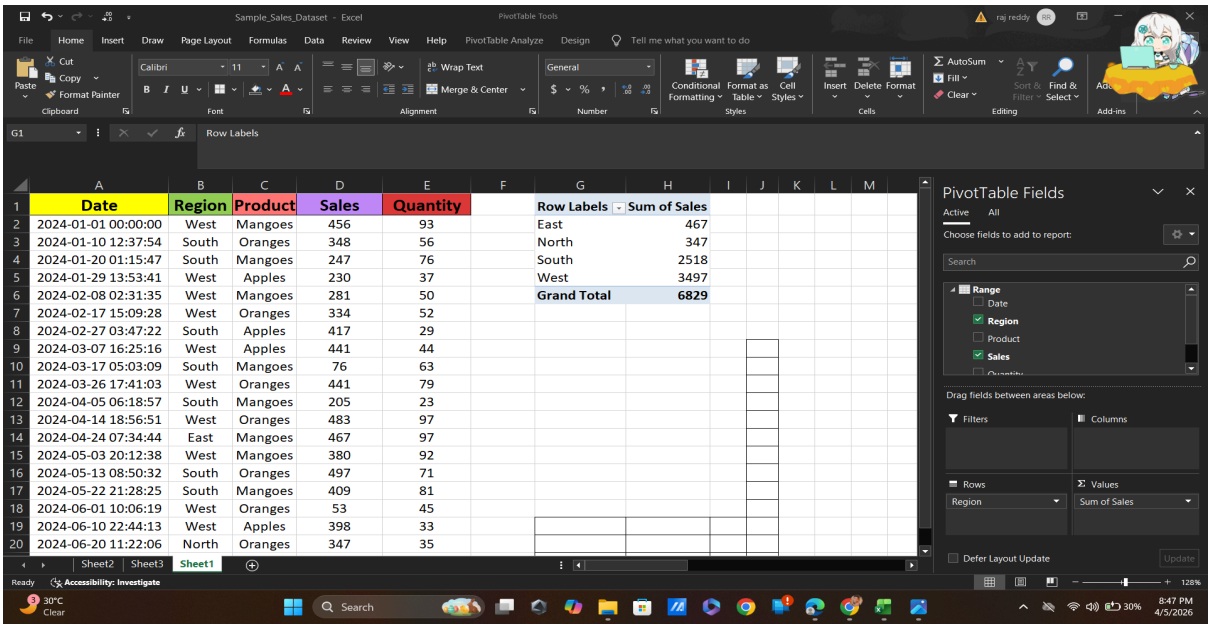
1. **Select Your Data:** Highlight the table containing your sales data, ensuring it has clear headers (e.g., Region, Sales Amount).
2. **Insert PivotTable:** Go to the Insert tab on the Excel ribbon and click PivotTable.
3. **Choose Location:** In the dialog box, select New Worksheet and click OK.
4. Configure Fields: In the PivotTable Fields pane on the right:
    1. **Drag Region to the Rows area.**
    2. **Drag Sales Amount (or Revenue) to the Values area.**
5. **Verify/Adjust:** Ensure the values area displays "Sum of Sales Amount". If it shows "Count", right-click a value, select Summarize Values By > Sum.

6. **Formatting:** To format as currency, right-click any number, select Number Format, and choose "Currency" or "Accounting".

Customization Options

- **Sorting:** Right-click on a region name in the table and choose Sort to order from highest to lowest sales.
- **Filtering:** Add a Slicer via the *PivotTable Analyze* tab for quick filtering by region.
- **Refresh:** If raw data changes, go to PivotTable Analyze > Refresh to update the report.

Example of Pivot Table:



Date	Region	Product	Sales	Quantity
2024-01-01 00:00:00	West	Mangoes	456	93
2024-01-10 12:37:54	South	Oranges	348	56
2024-01-20 01:15:47	South	Mangoes	247	76
2024-01-29 13:53:41	West	Apples	230	37
2024-02-08 02:31:35	West	Mangoes	281	50
2024-02-17 15:09:28	West	Oranges	334	52
2024-02-27 03:47:22	South	Apples	417	29
2024-03-07 16:25:16	West	Apples	441	44
2024-03-17 05:03:09	South	Mangoes	76	63
2024-03-26 17:41:03	West	Oranges	441	79
2024-04-05 06:18:57	South	Mangoes	205	23
2024-04-14 18:56:51	West	Oranges	483	97
2024-04-24 07:34:44	East	Mangoes	467	97
2024-05-03 20:12:38	West	Mangoes	380	92
2024-05-13 08:50:32	South	Oranges	497	71
2024-05-22 21:28:25	South	Mangoes	409	81
2024-06-01 10:06:19	West	Oranges	53	45
2024-06-10 22:44:13	West	Apples	398	33
2024-06-20 11:22:06	North	Oranges	347	35

Row Labels	Sum of Sales
East	467
North	347
South	2518
West	3497
Grand Total	6829

3. Apply Conditional Formatting to highlight sales greater than 400

Ans-

Steps for Excel:

1. **Select the Data:** Highlight the range of cells (e.g., column of sales figures) that you want to apply the formatting to.
2. **Open Conditional Formatting:**
    1. Go to the Home tab in the ribbon.
    2. Click on Conditional Formatting in the Styles group.
3. **Create the Rule:**
    1. Select Highlight Cells Rules.
    2. Choose Greater Than400

4. **Set the Threshold and Format:**

    1. In the dialog box, type `400` in the value box.

    2. In the dropdown menu to the right, select a formatting style (e.g., "Light Red Fill with Dark Red Text" or create a custom format).

4. **Finish:** Click OK.

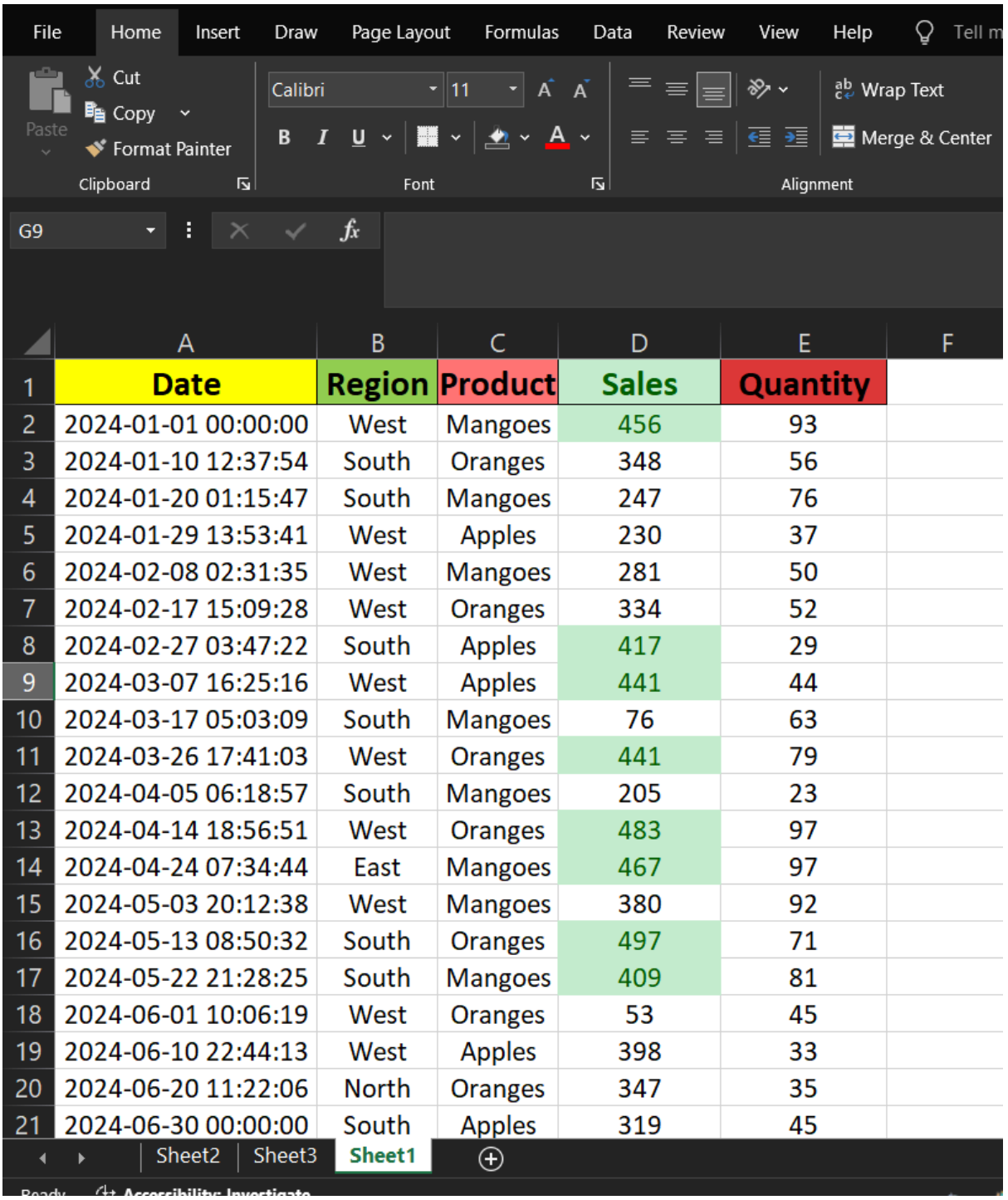
Alternative Method (Using "Quick Analysis")

1. Select the sales data.
2. Click the Quick Analysis button that appears in the bottom-right corner of your selection (or press `Ctrl+Q`).
3. Choose Greater Than, enter `400`, and click OK.

Pro-Tips

- Dynamic Updating: If you change a value to be less than 400, the formatting will automatically disappear.

- **Managing Rules:** If you need to change the color or the threshold later, go to Conditional Formatting > Manage Rules > Edit Rule.
- **Highlight Entire Row:** To highlight the entire row based on a value in a specific column, use New Rule > Use a formula to determine which cells to format, and use a formula like `=$C1>400` (assuming column C has the sales data)
- **Example of Greater than 400**



Date	Region	Product	Sales	Quantity
2024-01-01 00:00:00	West	Mangoes	456	93
2024-01-10 12:37:54	South	Oranges	348	56
2024-01-20 01:15:47	South	Mangoes	247	76
2024-01-29 13:53:41	West	Apples	230	37
2024-02-08 02:31:35	West	Mangoes	281	50
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2024-04-05 06:18:57	South	Mangoes	205	23
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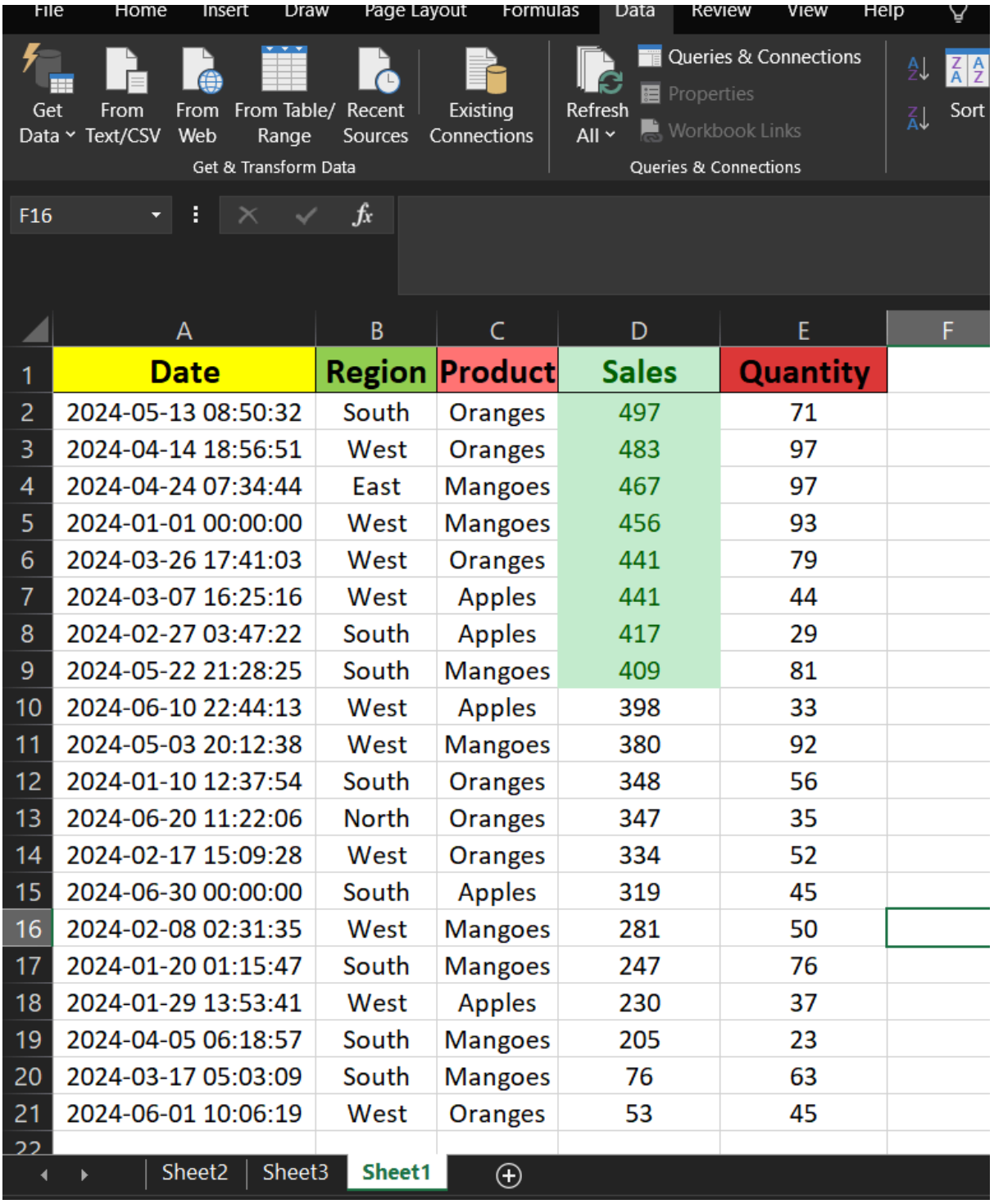
2024-05-03 20:12:38	West	Mangoes	380	92
2024-05-13 08:50:32	South	Oranges	497	71
2024-05-22 21:28:25	South	Mangoes	409	81
2024-06-01 10:06:19	West	Oranges	53	45
2024-06-10 22:44:13	West	Apples	398	33
2024-06-20 11:22:06	North	Oranges	347	35
2024-06-30 00:00:00	South	Apples	319	45

4. Explain the difference between Sorting and Filtering with an example from this dataset

Ans-

- **Sorting** rearranges the entire dataset into a specific order numerical, or chronological). All data remains visible; only its position changes.

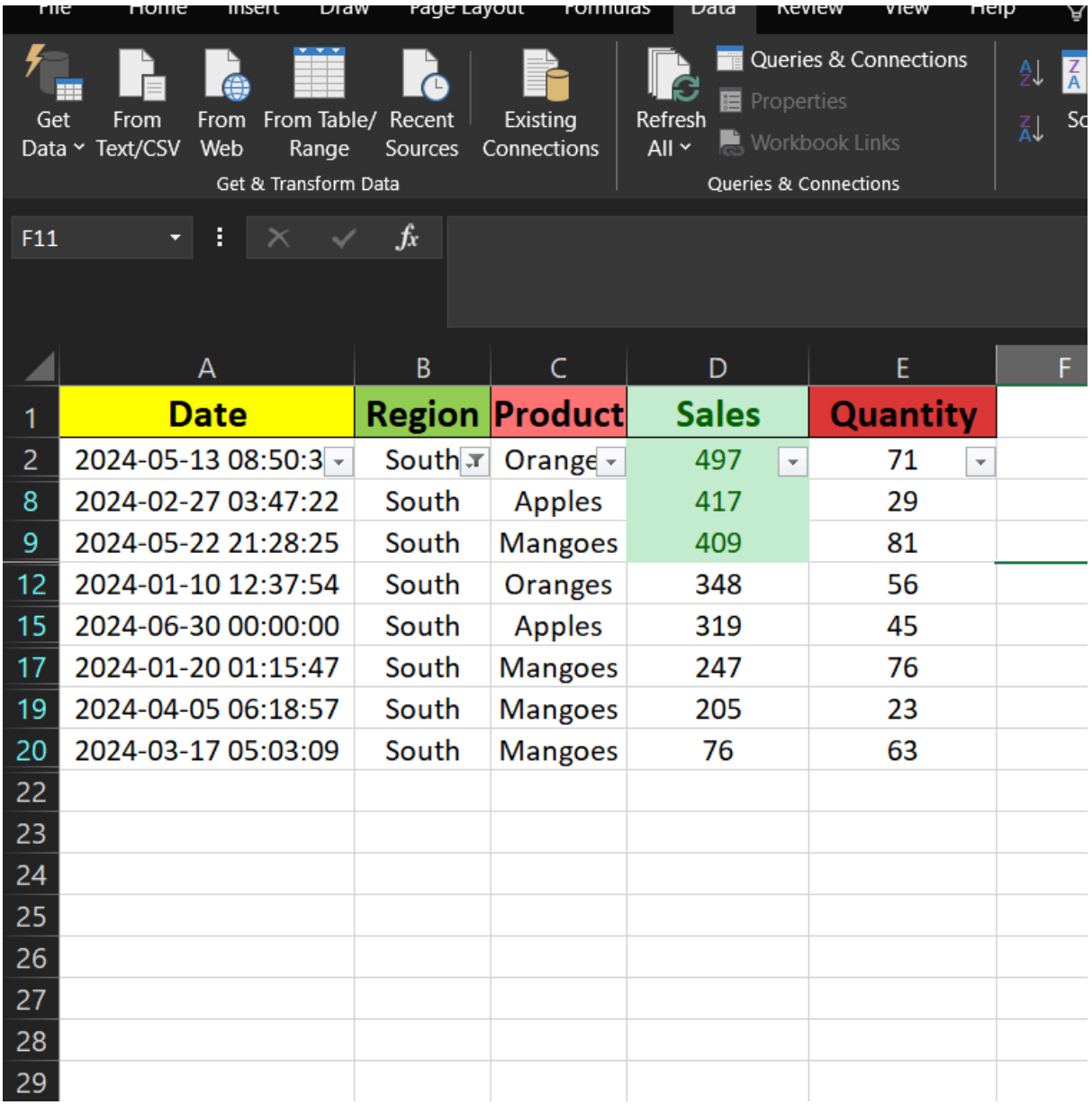
Action: Sort by **Sales-Descending**



	A	B	C	D	E	F
1	Date	Region	Product	Sales	Quantity	
2	2024-05-13 08:50:32	South	Oranges	497	71	
3	2024-04-14 18:56:51	West	Oranges	483	97	
4	2024-04-24 07:34:44	East	Mangoes	467	97	
5	2024-01-01 00:00:00	West	Mangoes	456	93	
6	2024-03-26 17:41:03	West	Oranges	441	79	
7	2024-03-07 16:25:16	West	Apples	441	44	
8	2024-02-27 03:47:22	South	Apples	417	29	
9	2024-05-22 21:28:25	South	Mangoes	409	81	
10	2024-06-10 22:44:13	West	Apples	398	33	
11	2024-05-03 20:12:38	West	Mangoes	380	92	
12	2024-01-10 12:37:54	South	Oranges	348	56	
13	2024-06-20 11:22:06	North	Oranges	347	35	
14	2024-02-17 15:09:28	West	Oranges	334	52	
15	2024-06-30 00:00:00	South	Apples	319	45	
16	2024-02-08 02:31:35	West	Mangoes	281	50	
17	2024-01-20 01:15:47	South	Mangoes	247	76	
18	2024-01-29 13:53:41	West	Apples	230	37	
19	2024-04-05 06:18:57	South	Mangoes	205	23	
20	2024-03-17 05:03:09	South	Mangoes	76	63	
21	2024-06-01 10:06:19	West	Oranges	53	45	

- **Filtering** selects a specific subset of data that meets certain criteria and hides the rest. It does not change the order of the original data; it only changes what is visible

Action:Filter **Region="South"**



	A	B	C	D	E	F
1	Date	Region	Product	Sales	Quantity	
2	2024-05-13 08:50:3	South	Orange	497	71	
8	2024-02-27 03:47:22	South	Apples	417	29	
9	2024-05-22 21:28:25	South	Mangoes	409	81	
12	2024-01-10 12:37:54	South	Oranges	348	56	
15	2024-06-30 00:00:00	South	Apples	319	45	
17	2024-01-20 01:15:47	South	Mangoes	247	76	
19	2024-04-05 06:18:57	South	Mangoes	205	23	
20	2024-03-17 05:03:09	South	Mangoes	76	63	
22						
23						
24						
25						
26						
27						
28						
29						

5. Create a PivotChart showing Monthly Sales Trend.

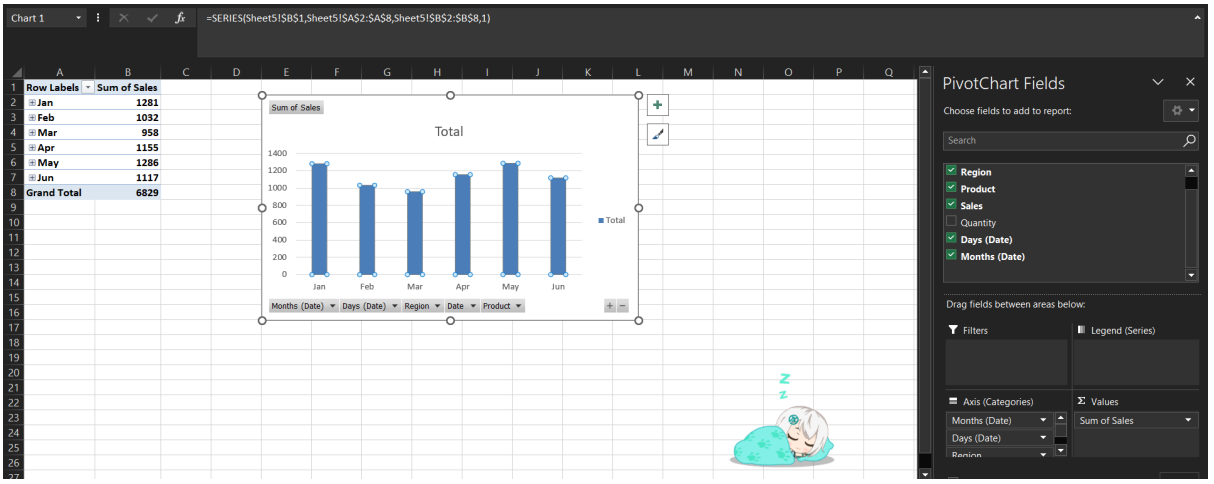
Ans-

Step-by-Step Instructions

1. **Prepare Your Data:**
   - Ensure your data is organized in a table format with clear headers (e.g., "Date", "Sales Amount").
   - Ensure there are no empty rows or columns.

- It is highly recommended to convert your data range into an official Excel Table by pressing `Ctrl + T`. This makes the PivotChart update automatically when new data is added.

2. **Insert PivotChart:**
   - Click any cell inside your data table.
   - Navigate to the Insert tab on the ribbon.
   - Click on PivotChart (or PivotTable -> PivotChart).
   - Choose to place the chart in a New Worksheet and click OK.
3. **Build the PivotTable/Chart:**
   - In the "PivotTable Fields" pane, drag the Date field into the Axis (Rows) area.
     - *Note:* Excel often automatically groups dates into Months. If it does not, right-click any date in the pivot table, select Group, and choose Months.
   - Drag the Sales Amount (or Revenue) field into the Values area.
   - Ensure the Value Field Setting is set to Sum.
4. **Format the PivotChart:**
   - Click on the chart and select a Line Chart to best display the trend over time.
   - Right-click on any field button in the chart and select "Hide All Field Buttons on Chart" for a cleaner look.
   - Add a chart title (e.g., "Monthly Sales Trend") and format axis titles for clarity



6. Use a Slicer to filter data by Product and observe changes in sales.

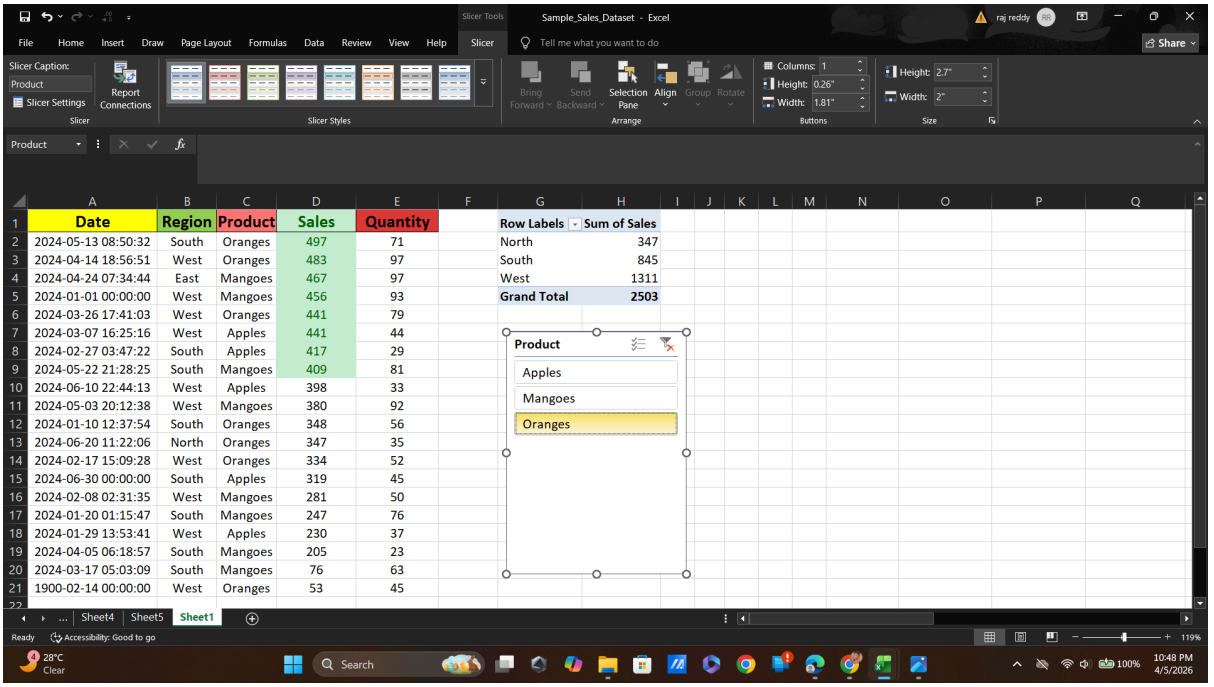
Ans-

Steps to Filter Data by Product and Observe Sales

1. **Create a PivotTable:** Select your sales data range, go to the Insert tab, and click PivotTable.
2. **Set Up the Sales View:** In the PivotTable Field List, drag "Product" to the Rows area and "Sales" to the Values area.
3. **Insert the Slicer:**
    1. **Click anywhere inside your PivotTable.**
    2. **Go to the PivotTable Analyze tab (or Analyze tab).**
    3. **Click Insert Slicer.**
4. **Select Product:** In the Insert Slicers dialog box, check the box for Product and click OK**.**
5. **Filter and Observe:**
    1. **A slicer box listing all products will appear.**
    2. **Click on a product button in the slicer to filter the PivotTable.**
    3. **Observation:** The PivotTable will instantly update to show the sales figures only for the selected product

.

4. **Multi-Select:** To compare multiple products, hold down the `Ctrl` key and click on multiple items in the slicer.
5. **Clear Filter:** Click the filter icon in the top-right corner of the slicer to return to viewing total sales.



7. Why are Dashboards important for decision-making in business analysis?
Ans-

- **Faster, Real-Time Decision-Making:** Dashboards provide instant access to data, removing the need for time-consuming, manual report creation and accelerating, for example, sales and logistics decisions.
- **Centralized "Single Source of Truth"**: They consolidate data from multiple sources (e.g., HR, Finance, Operations) into one place, aligning teams around the same KPIs and eliminating fragmented reporting.

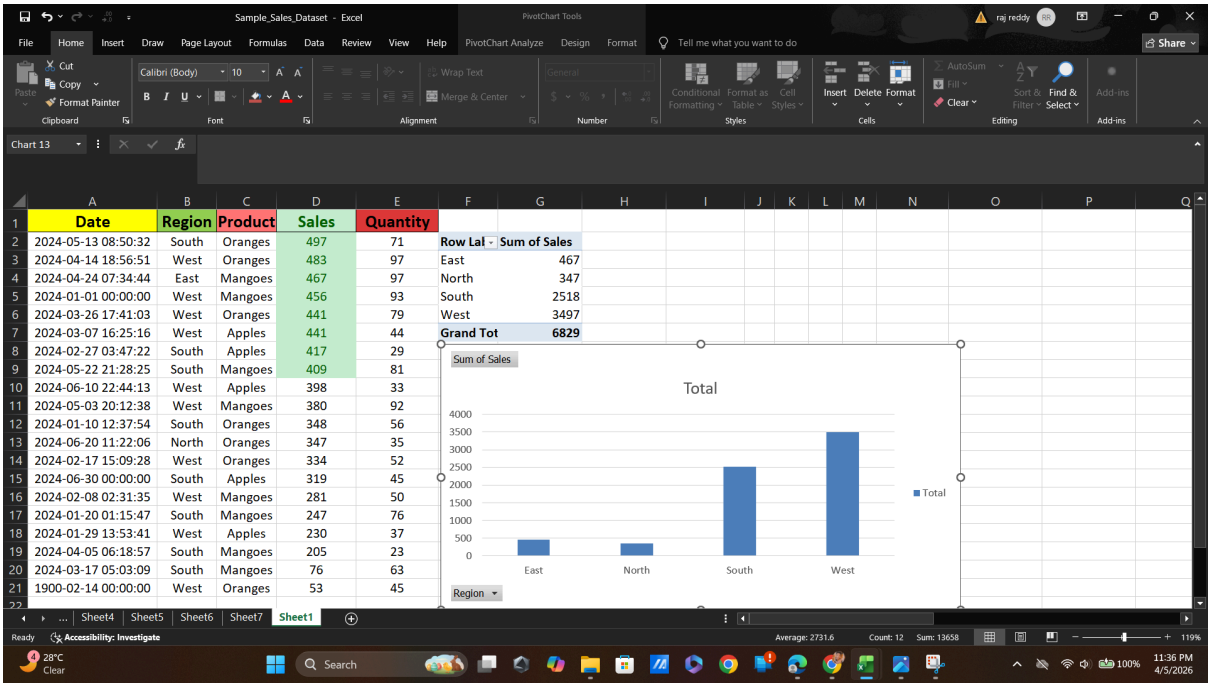
- **Improved Performance Monitoring:** By tracking metrics against targets in real-time, businesses can quickly detect problems and, for example, optimize manufacturing or marketing strategies.
- **Enhanced Forecasting & Analysis**: Dashboards allow users to identify trends, outliers, and patterns, facilitating better predictive analysis and strategic planning for the future.
- **Increased Operational Efficiency & Cost Savings**: By highlighting areas of waste or underperformance, businesses can optimize resource allocation and cut costs.

Core Components of Effective Dashboards:

- Interactive Visuals: Graphs, charts, and gauges transform complex data into actionable narratives.
- Customization: They can be tailored to specific user goals, roles, or departments, such as sales-specific insights.
- Key Performance Indicators (KPIs): Focused metrics provide a quick view of success, like for example, sustainability metrics.

8. Create a Dashboard that includes:

- Sales by Region (PivotTable + Chart)



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2024-04-14 18:56:51	West	Oranges	483	97
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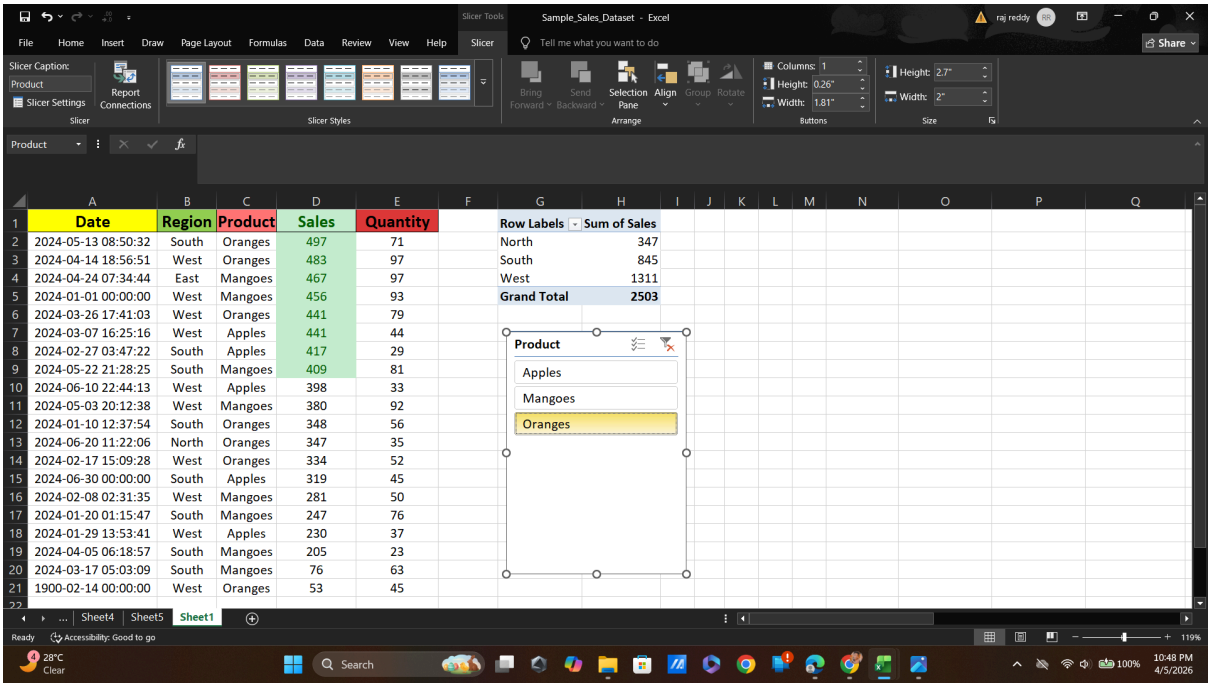
2024-04-05 06:18:57	South	Mangoes	205	23
2024-03-17 05:03:09	South	Mangoes	76	63
1900-02-14 00:00:00	West	Oranges	53	45

Row Lal	Sum of Sales
East	467
North	347
South	2518
West	3497
Grand Tot	6829

- **Product Filter (Slicer)**



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2024-04-14 18:56:51	West	Oranges	483	97
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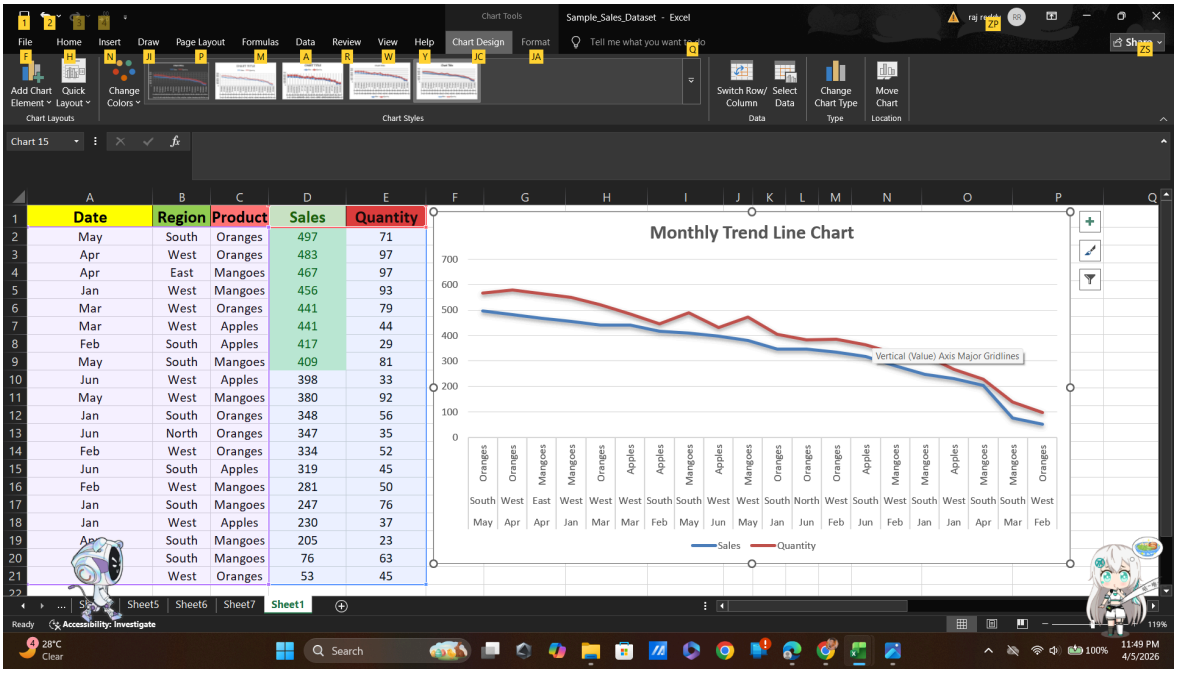
2024-03-17 05:03:09	South	Mangoes	76	63
1900-02-14 00:00:00	West	Oranges	53	45

Row Labels	Sum of Sales
North	347
South	845
West	1311
Grand Total	2503

- **Monthly Trend (Line Chart)**



Date	Region	Product	Sales	Quantity
May	South	Oranges	497	71
Apr	West	Oranges	483	97
Apr	East	Mangoes	467	97
Jan	West	Mangoes	456	93
Mar	West	Oranges	441	79
Mar	West	Apples	441	44
Feb	South	Apples	417	29
May	South	Mangoes	409	81
Jun	West	Apples	398	33
May	West	Mangoes	380	92
Jan	South	Oranges	348	56
Jun	North	Oranges	347	35
Feb	West	Oranges	334	52
Jun	South	Apples	319	45
Feb	West	Mangoes	281	50
Jan	South	Mangoes	247	76
Jan	West	Apples	230	37
Apr	South	Mangoes	205	23
Apr	South	Mangoes	76	63
Apr	West	Oranges	53	45

9. Group dates in the PivotTable into Months and calculate the average sales per month.

Ans-

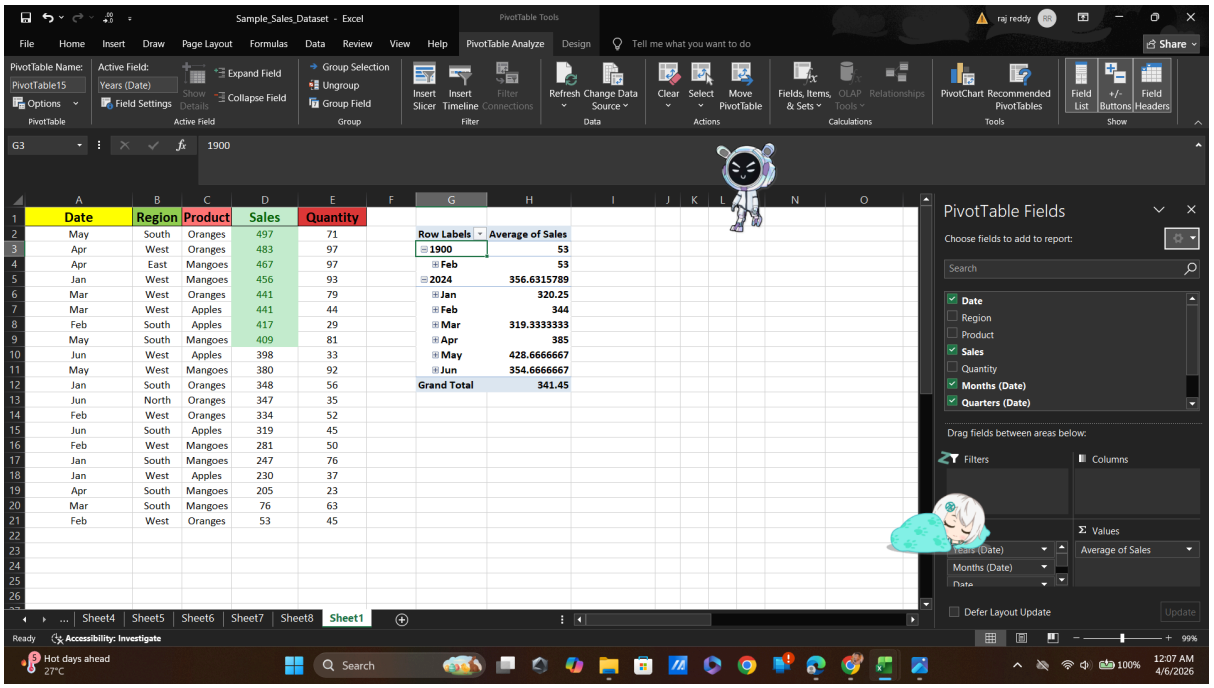
Detailed Steps:

1. **Create the PivotTable:** Select your data range and go to Insert > PivotTable.
2. **Add Fields:** Drag the Date field into the Rows area and the Sales field into the Values area.
3. **Group by Month:**
    1. Right-click any date in the Row Labels of the PivotTable.
    2. Select Group.
    3. In the dialog box, select Months. (Optional: select Years if your data spans multiple years to avoid combining January 2025 and January 2026).
    4. Click OK.
4. **Calculate Average:**
    1. In the Values area, click the arrow next to "Sum of Sales" and select Value Field Setting**s.**

2. Select Average as the summarization method.
3. Click OK.

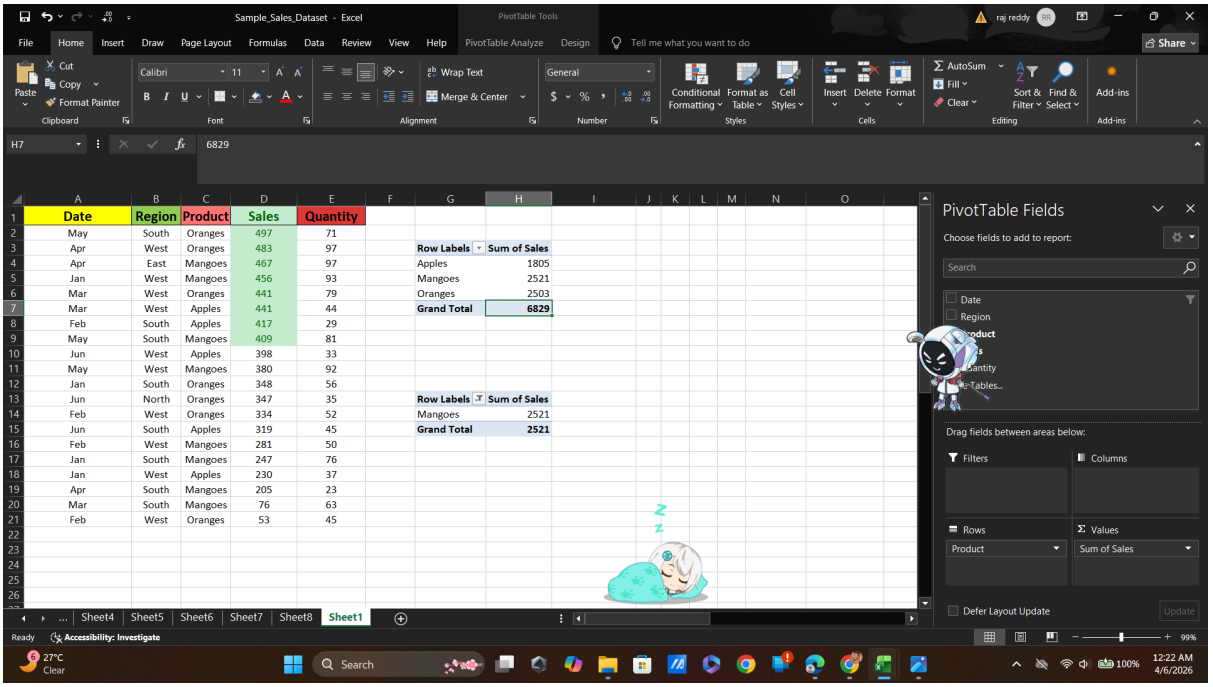
Tips:

- **Format Numbers**: You can format the resulting averages as currency by clicking "Number Format" in the Value Field Settings dialog.
- **Troubleshooting:** If grouping doesn't work, ensure all cells in your source date column are formatted as dates, not text



10. Combine Apples and Oranges into one group called Fruits and compare its sales with other products

Ans-


Sample_Sales_Dataset - Excel
PivotTable Tools
File
Home
Insert
Draw
Page Layout
Formulas
Data
Review
View
Help
PivotTable Analyze
Design
Tell me what you want to do
Share
Cut
Copy
Format Painter
Paste
Clipboard
Calibri
11
Font
Wrap Text
Merge & Center
Alignment
General
Number
Conditional Formatting
Format as Table
Cell Styles
Styles
Insert
Delete
Format
Cells
AutoSum
Fill
Clear
Sort & Filter
Find & Select
Editing
Add-ins
H7
6829
Date
Region
Product
Sales
Quantity
May South Oranges 497 71
Apr West Oranges 483 97
Apr East Mangoes 467 97
Jan West Mangoes 456 93
Mar West Oranges 441 79
Mar West Apples 441 44
Feb South Apples 417 29
May South Mangoes 409 81
Jun West Apples 398 33
May West Mangoes 380 92
Jan South Oranges 348 56
Jun North Oranges 347 35
Feb West Oranges 334 52
Jun South Apples 319 45
Feb West Mangoes 281 50
Jan South Mangoes 247 76
Jan West Apples 230 37
Apr South Mangoes 205 23
Mar South Mangoes 76 63
Feb West Oranges 53 45
Row Labels Sum of Sales
Apples 1805
Mangoes 2521
Grand Total 6829
Row Labels Sum of Sales
Mangoes 2521
Grand Total 2521
Sheet4
Sheet5
Sheet6
Sheet7
Sheet8
Sheet1
PivotTable Fields
Choose fields to add to report:
Search
Date
Region
Product
Sales
Quantity
More Tables...
Drag fields between areas below:
Filters
Columns
Rows
Values
Product
Sum of Sales
Defer Layout Update
Update
Ready
Accessibility: Investigate
99%
27°C
Clear
Search
100%
12:22 AM
4/6/2026